Charalambos Hadjipanayi, PhD

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EDUCATION

09/2020 – 02/2025 **Imperial College London - Department of Electrical and Electronic Engineering**
Doctor of Philosophy (Ph.D.)

Thesis: "Remote Gait Analysis for Assisted Living using Radar-on-chip Technology".
Developed and validated contactless radar-based methods for extracting clinically relevant spatiotemporal gait parameters, with validation in healthy participants and neurological populations.

Award:
Best Live Demonstration, IEEE BioCAS 2024

10/2016 - 06/2020 **Imperial College London - Department of Bioengineering**
Master of Engineering (M.Eng.) - Grade: 80.91% (1st Class)

Thesis: "Cybathlon 2020 Powered Exoskeleton Race".
Developed closed-loop sensory feedback for lower-limb exoskeleton control using plantar-pressure measurements.

Awards:
Dean's List (2017–2019; grades: 84.3%, 86.2%, 80.1%)

ENGINEERING EXPERIENCE

07/2024 - Present *Research Associate in Radar Signal Processing, Department of Electrical and Electronic Engineering, Imperial College London*

- Developing advanced radar-based remote sensing solutions for in-home monitoring of individuals with neurodegenerative disorders.
- Focusing on algorithm development, validation trials, and translational deployment of technology.

07/2019 - 09/2019 *Summer intern - Department of Electrical and Computer Engineering, University of Cyprus*

- Radar-based vital-sign extraction (ECG-validated) under supervision of Prof. Constantinos Pitris.

07/2018 - 08/2018 *Assistant Technician -* Scientronics Ltd. Cyprus

- Medical device installation and maintenance.

TEACHING EXPERIENCE

02/2021 - 09/2024 *Graduate Teaching Assistant Bioengineering Department, Imperial College London.*

- Supervised laboratories and assessed courseworks/examination.
- Head teaching assistant for "Digital Biosignal Processing" module in 2024.

SKILLS

Programming: MATLAB, Python, C, C++
Software: Solidworks, Autodesk Fusion 360, Vicon Nexus
Languages: English, Greek (fluent)

PUBLICATIONS

(a) *Peer-reviewed*

- **C. Hadjipanayi** *et al.*, “Remote Gait Analysis Using Ultra-Wideband Radar Technology Based on Joint Range-Doppler-Time Representation,” *IEEE Trans. Biomed. Eng.*, 2024.
- **C. Hadjipanayi,** M. Yin, and T. G. Constandinou, “Capturing gait parameters during asymmetric overground walking using ultra-wideband radars: A preliminary study,” in *Proc. Int. Conf. IEEE Eng. Med. Biol. Soc. (EMBC)*, 2024.
- M. Yin**, C. Hadjipanayi,** *et al.*, “Towards unobtrusive sleep health assessment using impulse radar: A pilot study in an elderly cohort,” *IEEE Trans. Biomed. Eng.* (submitted).
- M. Yin, **C. Hadjipanayi,** *et al.*, “Human gait identification using UWB radar micro-Doppler signature”, in *Proc. Int. Conf. IEEE Eng. Med. Biol. Soc. (EMBC).*
- M. Crook-Rumsey, S. J. C. Daniels, S. Abulikemu, H. Lai, A. Rapeaux, **C. Hadjipanayi,** *et al.*, “Multicohort cross-sectional study of cognitive and behavioural digital biomarkers in neurodegeneration: the Living Lab study protocol,” in *Bmj Open*, 2023.
- Z. Chen, **C. Hadjipanayi,** *et al.*, “Live Demonstration: Real-Time Gait and Respiration Analysis using Ultra-Wideband Radar,” in Proc. IEEE Biomed. Circuits Syst. (BIOCAS), 2024.
- A. Bannon, A. Rapeaux, O. Savolainen, **C. Hadjipanayi**, M. Yin, C. J. Chan, "Enabling Low-Cost Ultra-High-Framerate WiFi-Networked UWB Radar Systems for Indoor Monitoring of Physiology and Activity," *in Proc. IEEE Int. Radar Conf. (RADAR)*, 2025.

(b) *Under Review*

- J. J. Yun, **C. Hadjipanayi**, A. Jahangiri, A. Bannon, T. G. Constandinou, S. Haar, “Passive Sensing of Gait and Medication-related Fluctuations in Parkinson's Disease,” *medRxiv preprint*, 2025, doi: 10.1101/2025.11.12.25340068.
- **C. Hadjipanayi**, M. Yin, A. Bannon, Z. Chen, T. G. Constandinou, “Towards Radar-Agnostic Gait Analysis Across UWB and FMCW Systems”, *arXiv preprint*, 2026, doi: 10.48550/arXiv.2601.04415.
- M. Yin, K. G. Ravindran, **C. Hadjipanayi**, A. Bannon, A. Rapeaux, C. Della Monica, T. S. Lande, Derk-Jan Dijk, T. G. Constandinou, “Using Legacy Polysomnography Data to Train a Radar System to Quantify Sleep in Older Adults and People living with Dementia”, *arXiv preprint*, 2026, doi: 10.48550/arXiv.2601.04057.

PATENT

Radar System and Associated Methods

- International patent application: WO2025/229322 (PCT/GB2025/050916)
- Priority application: GB 2406092.3 (filed 30 April 2024)
- Inventors (equal): T. Constandinou, N. Yogendran, A. Bannon, A. Rapeaux, **C. Hadjipanayi**, Maowen Yin
- Publication date: 6 November 2025